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TESI DI LAUREA TRIENNALE

Numerical Simulations of Vortex Dynamics in Quantum Fluids

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Le matrici sono pecore travestite da lupi ~ Leonard Peter Bos

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Contents

Chapter 1

Introduction

1.1. Introduction to superfluidity

The equations of classical fluid dynamics, such as the Navier-Stokes equations, have been known for more than a century. However, the complete understanding of their solutions in a turbulent regime remains one of the most important unsolved problems in classical physics; in particular the statistical and geometric structure of turbulence is still a major topic of current research (see e.g. Frisch's review of Kolmogorov theory [3]).

With that said, an apparently unrelated topic is quantum mechanics. Bosons are quantum entities that can simultaneusly occupy the same state, unlike fermions because of the Pauli exclusion principle. When a large number of bosons are cooled to temperatures near absolute zero, an intriguing phenomenon occurs: a condensate forms and superfluidity emerges as a macroscopic, coherent quantum state [10]. In such a state, virtually all particles occupy the same quantum state, allowing flow without viscosity and with quantized vortices.

The superfluid can be described by a single macroscopic wave function, which encodes both density and phase (thus velocity) information. Its time evolution is governed by the Gross-Pitaevskii equation (GPE for short), a nonlinear Schrödinger-type equation that accounts for interactions among particles [4, 9]. Unlike linear Schrödinger evolution, the nonlinearity in GPE allows phenomena such as vortex cores-regions of near-zero density-around which circulation is quantized.

Despite its quantum nature, the superfluid often exhibits behavior that parallels classical fluid phenomena: quantized vortices move, interact, and can even reconnect, producing structures reminiscent of vortex filaments in classical turbulence [2].

These events though, makes the GPE analitically almost intractable, especially in the presence of multiple vortices or reconnection events. Therefore, it has been widely studied through numerical simulations, ([12, 6, 7] as just a few examples).

The aim of this thesis is, therefore, to study in detail an efficient numerical method for solving the GPE in the presence of straight vortices, both from the perspective of numerical analysis and via computational experiments illustrating some examples of straight vortex dynamics. We will employ a time-splitting method combined with finite differences on both uniform and non-uniform grid, which allows for high spatial resolution near the vortex core while keeping the computational cost manageable.

1.2. The mathematical context

1.2.1. The Gross-Pitaevskii equation

The governing equation for a weakly interacting Bose-Einstein condendate is, as mentioned before, the *Gross-Pitaevskii equation*

$$\mathrm{i}\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\Delta\psi + V_0 |\psi|^2 \psi - E_0\psi$$

where $\psi = \psi(\mathbf{r}, t)$ is the complex macroscopic wavefunction for a condensate of N bosons of mass m at position $\mathbf{r}$ and time t. The constant $\hbar$ is the reduced Planck's constant, V_0 is the strength of the repulsive interactions between bosons, and E_0 is the chemical potential of the system (the energy required to add a boson to the condensate).

The adimensional form of the GPE, whose derivation can be found in [12], is

$$\frac{\partial \psi}{\partial t} = \frac{\mathrm{i}}{2}\Delta\psi + \frac{\mathrm{i}}{2}(1 - |\psi|^2)\psi \tag{1.1}$$

1.2.2. Classical analogy via the Madelung transformation

To give an idea of the analogy with classical fluid mechanics, as previously mentioned, we will give a brief overview of how the GPE can be rearranged to obtain a system of equations that closely resembles the Navier-Stokes equations for an inviscid and incompressible flow.

This is achieved through the *Madelung transformation* [8], which expresses the complex wavefunction ψ in terms of a real-valued density ρ and a phase S as follows:

$$\psi(\mathbf{r}, t) = \rho^{1/2}\exp(\mathrm{i}S) \tag{1.2}$$

where $\rho = \rho(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2$ is the density and $S = S(\mathbf{r}, t)$ is the phase of ψ. The vector field $\mathbf{u} = \nabla S$ can be interpreted as the superfluid velocity field.

Substituting (1.2) into (1.1) and separating real and imaginary parts, after some straightforward algebra [12], we obtain the following system of equations, written in tensorial notation

$$\begin{aligned} &\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_j)}{\partial x_j} = 0 \\ &\rho\left(\frac{\partial u_i}{\partial t} + u_j\frac{\partial u_i}{\partial x_j}\right) = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j}, \quad i = 1, 2, 3 \end{aligned}$$

where

$$p = \frac{\rho^2}{4} \quad \text{and} \quad \tau_{ij} = \frac{1}{4}\rho\frac{\partial^2 \log\rho}{\partial x_i x_j}$$

are the pressure and the quantum stress tensor, respectively.

We can clearly recognize the mass continuity equation in the first equation, being formally identical to its classical counterpart. Differences begin to emerge in the second equation, which essentially expresses the conservation of momentum.

The quantum fluid is inherently *inviscid*. Unlike the viscous stress tensor d_{ij} [11] present in the Navier-Stokes equations, which is proportional to the velocity gradient and dissipates kinetic energy through friction, while the *quantum stress tensor* τ_{ij} introduces no dissipation. Instead, this term redistribute the energy within the system [13].

These deep formal similarities suggest that the study of quantum vortex dynamics, in particular *non-dissipative vortex reconnection* events, could provide valuable insights into understanding the complex structure of turbulence in the classical case.

1.3. Approximation of steady-state vortex

In what follows, we will focus on the two-dimensional case, which is sufficient to describe straight vortices in three dimensions. Here we will seek an expression for the density profile in the case of a single steady-state vortex, centered at the origin. Let us start by imposing that and the time independency. In a 2-dimensional domain, with these constraints, the wavefunction has the form

$$\psi(x, y) = f(r) \exp\left(\mathrm{i}\,\theta(x, y)\right)$$

where $r = \sqrt{x^2 + y^2}$ and $f(r) = \sqrt{\rho(r)}$.

Now, we need to determine the function $f(r)$. By imposing the time independency condition, and substituting the above expression into the GPE (1.1), we obtain the following boundary value problem:

$$f'' + \frac{f'}{r} + f\left(1 - f^2 - \frac{1}{r^2}\right) = 0 \tag{1.3}$$

with boundary conditions $f(0) = 0$ and $f(\infty) = 1$. There are several approaches: one could solve the boundary-value problem directly on a truncated domain and interpolate the solution. In many applications, however, it is convenient to approximate $f(r)$ using a Padé approximant [1], in order to be able to evaluate the function everywhere. In what follows we will use a 4-th order Padé approximant, which coefficients can be found in [6].

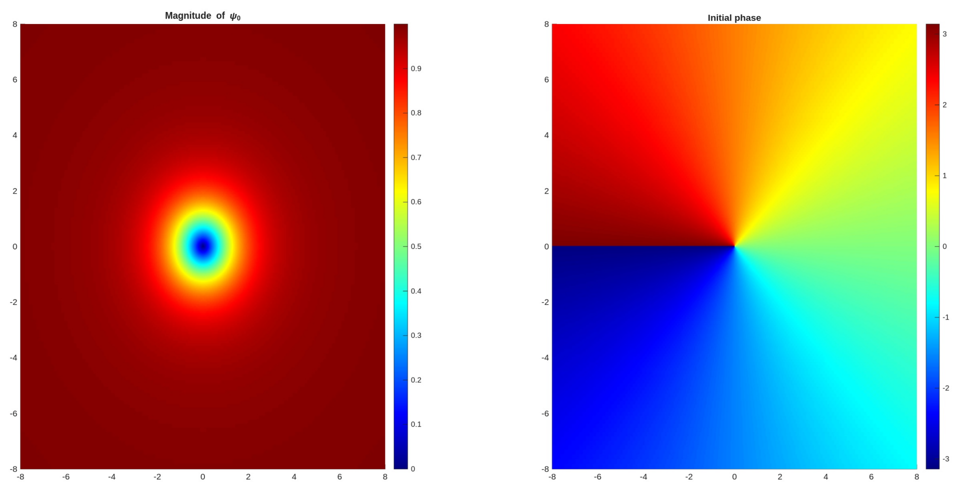

Figure 1.1.: Magnitude $|\psi_0| = \sqrt{\rho}$ (left), initial phase in radiants (right)

CHAPTER 2

NUMERICAL METHODS

2.1. Space discretization

In this section we describe the spatial discretization methods used to simulate the GPE.

The choice of an appropriate spatial discretization is crucial to ensure both the stability and the accuracy of the numerical scheme. Different approaches can be employed depending on the geometry of the problem and the desired level of precision; in our case, we focus on finite difference methods, both on uniform and non-uniform grids, which provide a straightforward and efficient way to approximate spatial derivatives.

The following subsections outline the adopted discretization strategy and discuss its main numerical properties.

2.1.1. Finite differences on uniform grid

Let us discretize the interval $[-L, L]$ with m equispaced points, so that

$$x_1 = -L, \quad x_m = L, \quad x_j = -L + (j-1)h, \quad 1 \le j \le m,$$

where the step size is given by $h = \frac{2L}{m-1}$. Following [5], the second derivative along one spatial direction is approximated by the standard finite-difference matrix

$$D_2^x = \frac{1}{h^2}\begin{bmatrix} -2 & 2 & 0 & 0 & \cdots & 0 & 0 \\ 1 & -2 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 & 0 \\ 0 & 0 & 1 & -2 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & \cdots & 2 & -2 \end{bmatrix}$$

This corresponds to the usual second-order central difference scheme with homogeneous Neumann boundary conditions.

2.1.2. Finite differences on non-uniform grid

Although the above numerical scheme is simple and easy-to-implement, in our case it is not the most suitable one, since, as we will see in the next chapter, it requires a very large number of points to obtain physically reliable results, especially when integrating over long time intervals.

Given that the vortex density exhibits stronger variations near its core, while remaining almost constant towards the boundary of the domain, it would be preferable to employ a grid that clusters points around the vortex center(s). In this way, higher resolution is achieved in the regions where sharp variations occur.

Consider the interval $[a, b]$ and let $\mathbf{x} = [x_1, \ldots, x_m]^T$, with

$$x_1 = a, \quad x_m = b, \quad x_j = x_{j-1} + h_j,$$

where the grid spacings satisfy $h_j = (1+\delta)h_{j-1}$ for some fixed parameter δ. This leads to the recurrence

$$\begin{aligned} h_j &= (1+\delta)h_{j-1} \\ &= (1+\delta)^2 h_{j-2} \\ &\vdots \\ &= (1+\delta)^{j-1} h_1. \end{aligned}$$

The constraint is that the sum of all step sizes must cover the entire interval length, which gives

$$\sum_{j=1}^{m-1} h_j = h_1 \sum_{j=1}^{m-1} (1+\delta)^{j-1} = b - a.$$

There are three natural ways to proceed, each consisting of fixing two among the parameters δ, m, and h_1, and then computing the remaining one. In what follows, we present two cases: computing m and computing h_1.

First of all, we observe that the sum is a geometric series with ratio $1+\delta$, whose closed-form expression is known analytically:

$$\sum_{j=1}^{m-1} (1+\delta)^{j-1} = \frac{(1+\delta)^{m-1} - 1}{\delta}.$$

To compute m we proceed as follows:

$$\begin{aligned} h_1 \frac{(1+\delta)^{m-1} - 1}{\delta} = b - a \iff & (1+\delta)^{m-1} = \frac{\delta(b-a)}{h_1} + 1 \\ \iff & (m-1)\log(1+\delta) = \log\left(\frac{\delta(b-a)}{h_1} + 1\right) \\ \iff & m = \left\lceil \frac{\log\left(\frac{\delta(b-a)}{h_1} + 1\right)}{\log(1+\delta)} + 1 \right\rceil. \end{aligned}$$

Since m must be an integer, we take the ceiling in the final expression.

The expression for h_1, after some trivial algebra, is

$$h_1 = \frac{\delta(b-a)}{1 - (1+\delta)^{m-1}}$$

2.1.3. On the grid construction

Since we are truncating the domain, the error introduced by the artificial boundary conditions is not negligible. Naturally, considering a larger domain would improve the overall accuracy of the solution.

However, considering a uniform grid, enlarging the domain also requires increasing the number of points to achieve the same accuracy that we had on the smaller domain with fewer points. This can be problematic, since doubling the domain means we would need *at least* twice as many points.

Otherwise, if we consider the firt presented way tho construct a non-uniform grid, fixing δ and h_1, the grid will be *almost* identical in both the smaller and the larger domains (except near the boundaries), with restricting the latter to the smaller one. This particular construction allows us to achieve *the same* high resolution near the center of the vortex core while keeping the total number of points reasonably low.

To construct the one-dimensional discretization, it is important to note that, in the case of non-uniform finite differences, the local error can be shown to be proportional to $O(h_j^2)$ provided that the difference between h_j and h_{j-1} is of order $O(\max\{h_j^2, h_{j-1}^2\})$, therefore, care must be taken in constructing the grid using the above formula, so as to keep the difference between two consecutive steps as smooth as possible. Otherwise, the error would not even be proportional to h_j, which in the worst case would lead to an inconsistent scheme and consequently to significant errors in the time integration.

With this in mind, we can construct the discretization along one direction by applying the procedure described above to the interval $[0, x_c]$, where x_c denotes the center of the vortex, clustering the points towards x_c. We then extend the grid from x_c to the boundary of the domain, using the same values of h and δ, so that the spacing over the interval $[0, 2x_c]$ remains consistent with the previous construction. The grid can be further extended up to the domain boundary within a small tolerance, since this procedure does not necessarily guarantee that the endpoints are reached exactly. However, this is not problematic, as small discrepancies at the domain boundaries do not significantly affect the evolution of the dynamics, being far from the regions of interest.

Once we have the step vector $\mathbf{h}$ constructed, we can discretize the second derivative along one direction using the tridiagonal matrix

$$D_2^x = \begin{bmatrix} -\frac{2}{h_1^2} & \frac{2}{h_1^2} & 0 & \dots & 0 \\ \frac{2}{h_2(h_1+h_2)} & -\frac{2}{h_1 h_2} & \frac{2}{h_1(h_1+h_2)} & \dots & 0 \\ 0 & \frac{2}{h_3(h_2+h_3)} & -\frac{2}{h_2 h_3} & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \frac{2}{h_{m-1}(h_{m-2}+h_{m-1})} \\ 0 & \dots & 0 & \frac{2}{h_{m-1}^2} & -\frac{2}{h_{m-1}^2} \end{bmatrix}$$

2.1.4. Laplacian discretization

In both the above cases, the Laplacian operator in two dimension is approximated by

$$\Delta \approx I^y \otimes D_2^x + D_2^x \otimes I^x \tag{2.1}$$

where $\otimes$ denotes the Kronecker product (see A.1 for further details), and I^* denotes the identity matrix of the dimension equal to the number of points in each direction.

2.2. Time integration

2.2.1. Strang splitting method

Consider the Cauchy problem

$$\begin{cases} \frac{\partial u}{\partial t} = (A+B)u \\ u(0) = u_0 \end{cases} \tag{2.2}$$

where A and B are two differential operators.
The analytical solution for (2.2) is

$$u(t) = e^{(A+B)t}u_0$$

Proposition 2.1. *If A and B commute, then* $\exp(A+B) = \exp(A)\exp(B)$

Proof.

$$\begin{aligned}
\exp(A+B) &= \sum_{n=0}^{\infty} \frac{(A+B)^n}{n!} \\
&= \sum_{n=0}^{\infty} \sum_{k=0}^{n} \binom{n}{k} \frac{A^k B^{n-k}}{n!} \\
&= \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{A^k}{k!} \frac{B^{n-k}}{(n-k)!} \qquad \ell = n-k \\
&= \sum_{k=0}^{\infty} \frac{A^k}{k!} \sum_{\ell=0}^{\infty} \frac{B^\ell}{\ell!} \\
&= \exp(A)\exp(B)
\end{aligned}$$

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So, for the previous proposition, if A and B commute, then $u(t) = e^{tA}e^{tB}u_0 \implies u(t+\tau) = e^{\tau A}e^{\tau B}u(t)$.
This is equivalent to solve separately two initial value problems:

$$\begin{cases} \frac{\partial u}{\partial t} = Au \\ u(0) = u_0 \end{cases}$$

and

$$\begin{cases} \frac{\partial u}{\partial t} = Bu \\ u(0) = u_0 \end{cases}$$

This gives rise to a numerical scheme:

$$\begin{aligned}
\bar{u}_{n+1} &= \exp(\tau A)u_n \\
u_{n+1} &= \exp(\tau B)\bar{u}_{n+1}
\end{aligned}$$

This approach is known as *Lie splitting*. It is exact when the operators A and B commute, otherwise it achieves first-order accuracy with respect to the time step τ.

Other methods can be constructed by considering alternative ways of splitting the operators. One widely used approach is the *Strang time splitting*, which improves the accuracy by symmetrically composing the substeps. Specifically, it provides the numerical scheme

$$\begin{aligned} \bar{u}_{n+1} &= \exp\left(\frac{\tau}{2}A\right) u_n \\ \tilde{u}_{n+1} &= \exp(\tau B)\bar{u}_{n+1} \\ u_{n+1} &= \exp\left(\frac{\tau}{2}A\right) \tilde{u}_{n+1} \end{aligned} \tag{2.3}$$

where $u_n \approx u(t_n)$.

Proposition 2.2 (Error of Strang splitting)**.** *The Strang time-splitting method is* exact *if the operators A and B commute, and it is of order* 2 *with respect to the time step* τ *otherwise.*

Remark 2.3. In the special case where A and B commute, the Strang splitting becomes exact. Indeed, we have

$$e^{(A+B)\tau} = e^{\tau A}e^{\tau B} = e^{\frac{\tau}{2}A}e^{\frac{\tau}{2}A}e^{\tau B} = e^{\frac{\tau}{2}A}e^{\tau B}e^{\frac{\tau}{2}A},$$

which coincides with the splitting formulation.

Order 2. Let us now consider the general case where A and B do not commute. The exact solution after one time step is given by

$$u(t+\tau) = e^{(A+B)\tau}u(t),$$

whereas the Strang approximation reads

$$u(t+\tau) \approx e^{\frac{\tau}{2}A}e^{\tau B}e^{\frac{\tau}{2}A}u(t).$$

The difference between the two is entirely contained in the discrepancy between the operators

$$e^{(A+B)\tau} \quad \text{and} \quad e^{\frac{\tau}{2}A}e^{\tau B}e^{\frac{\tau}{2}A}.$$

To quantify this, we expand the Strang operator in a Taylor series:

$$\left(I + \tfrac{\tau}{2}A + \tfrac{\tau^2}{8}A^2 + O(\tau^3)\right)\left(I + \tau B + \tfrac{\tau^2}{2}B^2 + O(\tau^3)\right)\left(I + \tfrac{\tau}{2}A + \tfrac{\tau^2}{8}A^2 + O(\tau^3)\right).$$

Multiplying and collecting terms up to order τ^2, we obtain

$$I + \tau(A+B) + \tfrac{\tau^2}{2}(A+B)^2 + O(\tau^3).$$

On the other hand, the Taylor expansion of the exact propagator is

$$e^{(A+B)\tau} = I + \tau(A+B) + \tfrac{\tau^2}{2}(A+B)^2 + \tfrac{\tau^3}{3}(A+B)^3 + O(\tau^4).$$

Subtracting the two expressions shows that the leading term of the local error is of order τ^3. Hence, the method is (at least) second-order accurate.

To prove that it is *exactly* 2, we need to expand up to the third order and subtract. After some computation, one finds that the local error is

$$\tau^3\left(\frac{1}{12}[B,[B,A]] - \frac{1}{24}[A,[A,B]]\right) + O(\tau^4) = O(\tau^3)$$

Since, as $[A,B] = AB - BA \neq 0$, the cubic term does not cancel, hence the method has a local error proportional to τ^3, and is therefore of order 2.

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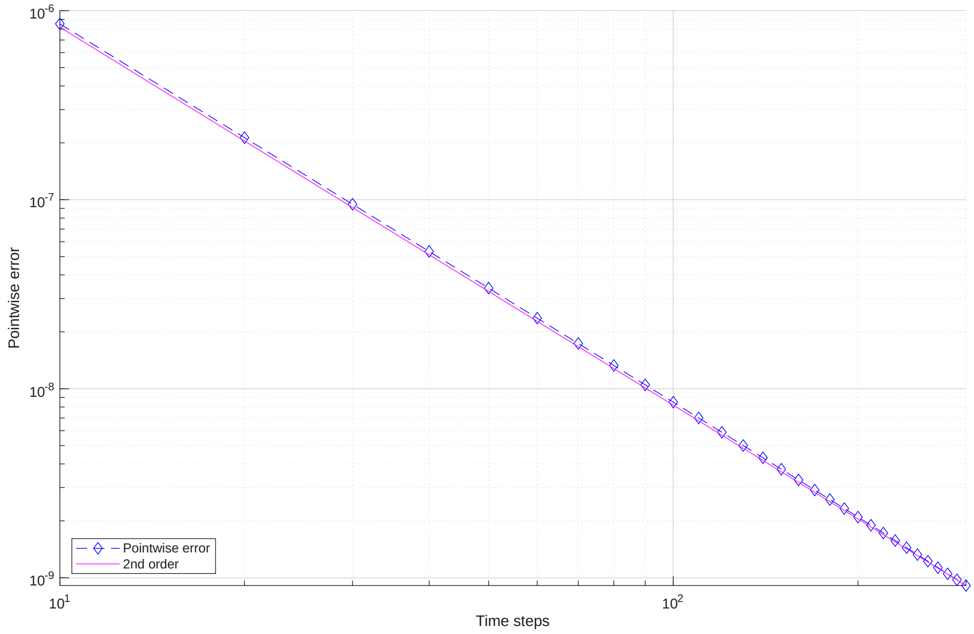

Figure 2.1.: Convergence order

2.2.2. Method applied to GPE

In the case of GPE, we have $A = \frac{i}{2}\Delta$ and $B = \frac{i}{2}\left(1 - |u|^2\right)$, this results in the numerical scheme

$$\bar{\mathbf{u}}_{n+1} = \exp\left(\frac{\tau i}{2}\Delta\right)\mathbf{u}_n \tag{2.4}$$

$$\tilde{\mathbf{u}}_{n+1} = \exp\left(\frac{\tau i}{2}\left(1 - |\bar{\mathbf{u}}_{n+1}|^2\right)\right)\bar{\mathbf{u}}_{n+1} \tag{2.5}$$

$$\mathbf{u}_{n+1} = \exp\left(\frac{\tau i}{2}\Delta\right)\tilde{\mathbf{u}}_{n+1} \tag{2.6}$$

For the non-linear part (2.5) we directly compute the exact solution as

$$\tilde{\mathbf{u}}_{n+1} = \exp\left(\mathrm{diag}\{\tau d_n\}\right)\bar{\mathbf{u}}_{n+1}$$

where $d_n = \frac{i}{2}\left(1 - |\bar{\mathbf{u}}_{n+1}|^2\right)$. The notation $\mathrm{diag}\{\mathbf{v}\}$, for a generic vector $\mathbf{v} \in \mathbb{R}^m$, denotes the diagonal matrix

$$\mathrm{diag}\{\mathbf{v}\} = \begin{bmatrix} v_1 & 0 & \cdots & \cdots & 0 \\ 0 & v_2 & 0 & \cdots & 0 \\ \vdots & 0 & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & v_m \end{bmatrix}$$

In this case, no matrix has to be created, since is enough to get the element-wise exponentiation of the vector τd_n.

The treatment for the Laplacian operator is more delicate: since in fact the latter is approximated by the formula (2.1), we need to compute the matrix-vector product

$$\exp\left(\frac{\tau}{2}\left(I^y \otimes D_2^x + D_2^y \otimes I^x\right)\right)\mathbf{u}_n$$

which turns out to be a full matrix-vector product of large dimension, leading to a huge memory requirement and, consequently, to an unreasonable computational time. Our goal would be avoiding the construction of any Kronecker product; instead we directly compute the action of the latter on the vector $\mathbf{u}$. First we notice that $I^y \otimes D_2^x$ and $D_2^y \otimes I^x$ commute, as trivial consequence of the formula (A.3). This implies that

$$\exp\left(I^y \otimes D_2^x + D_2^y \otimes I^x\right) = \exp\left(I^y \otimes D_2^x\right)\exp\left(D_2^y \otimes I^x\right)$$

Since the exponential function is analytical, using (A.8), we have

$$\exp\left(I^y \otimes D_2^x\right) = I^y \otimes \exp\left(D_2^x\right)$$

and the same for the other. Using again the formula (A.3), we obtain

$$\exp\left(I^y \otimes D_2^x + D_2^y \otimes I^x\right)\mathbf{u} = \left(\exp\left(D_2^y\right) \otimes \exp\left(D_2^x\right)\right)\mathbf{u}$$

This is way better than directly compute the full exponential of Kronecker product, but it still requires the computation of a large full matrix. This can be avoided, since we can take into account the matrix $U = (u_{i,j}) \in \mathbb{C}^{m_x \times m_y}$, with the relation $U_{i,j} = \mathbf{u}_{i+m_x(j-1)}$, also denoted with $\mathrm{vec}(U) = \mathbf{u}$. Let now $E_x = \exp\left(D_2^x\right)$ and $E_y = \exp\left(D_2^y\right)$. Thanks to (A.4), we have that

$$\left(\exp\left(D_2^y\right) \otimes \exp\left(D_2^x\right)\right)\mathbf{u} = \mathrm{vec}\left(E_x\, U\, E_y^T\right)$$

This result is significantly more efficient than the previous one, as now we only have to compute two small matrix exponentials and two matrix-matrix products.

Chapter 3

Numerical experiments

3.1. Stationary vortex preservation over long time integration

The first numerical experiment we consider concerns the preservation of the steady-state solution of (1.1). Since this solution is stationary, a natural way to assess the quality of our numerical method is to examine how much the approximate solution deviates from the initial one over long integration times.

In all the following experiments, we approximate the density ρ using a 4-th order Padé approximation of the equation (1.3)

From now on, the notation ψ_t denotes the numerical solution at time $t > 0$.

3.1.1. Uniform grid

We now present some results from the performed experiments. The parameters used in the following simulations are: $m_x = m_y = 347$, $n_t = 450$, $t^* = 20$, $L_x = L_y = 30$.

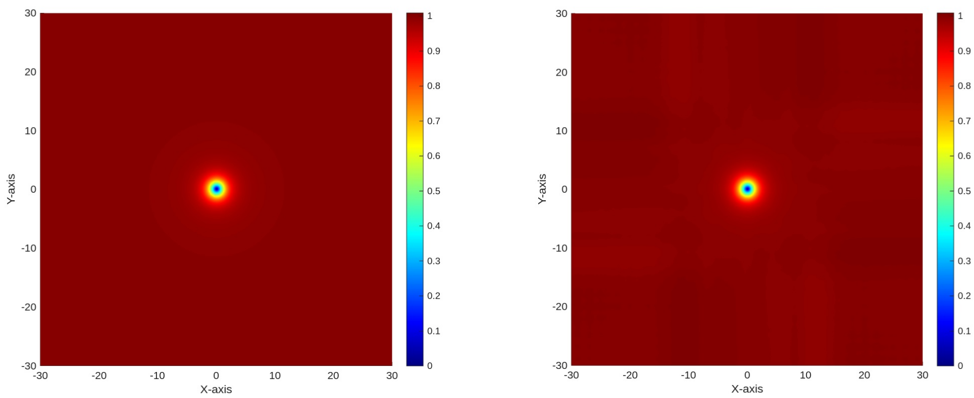

Figure 3.1.: $|\psi_0|$ (left) and $|\psi_{20}|$ (right), finite differences on a uniform grid.

In this figure, we can observe that the numerical solution integrated up to $t = 20$ shows slight variations in the norm outside the vortex core. To highlight this difference more clearly, in Figure 3.2 we employ an alternative colormap with a steeper gradient for values close to 1

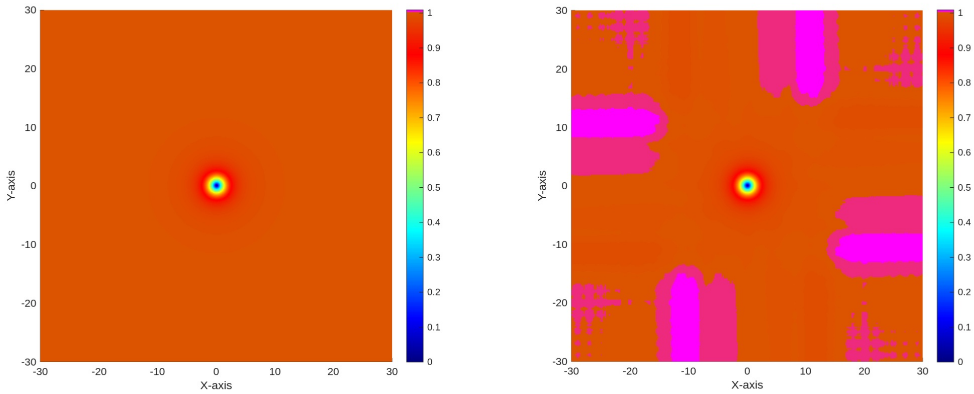

Figure 3.2.: Initial solution (left), solution at time $t = 20$ (right)

We observe that the variations in the norm are localized near the boundaries of the domain, outside the vortex core. To highlight this, Figure 3.3 shows the truncated domain $[-5, 5]$, emphasizing the vortex center.

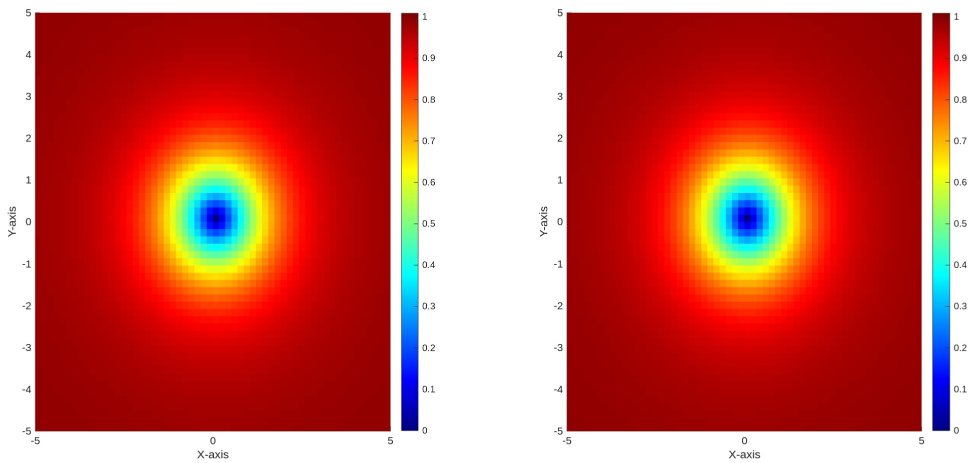

Figure 3.3.: Zoomed domain

3.1.2. Non-uniform grid

We now consider a non-uniformly spaced grid as described in Section 2.1.2, with $\delta = h_1 = 1/50$. This results in a grid composed of 347 points per direction.

We directly report the comparison between the uniform and non-uniform grids.

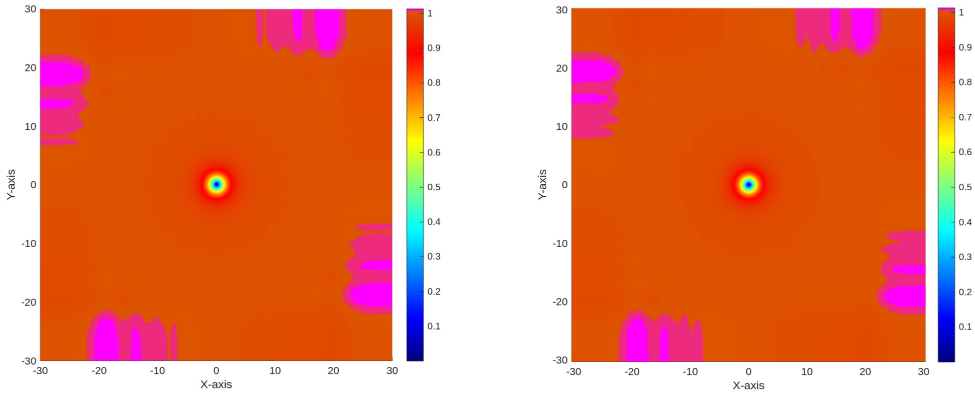

Figure 3.4.: Comparison between uniform (left) and non-uniform (right) grids.

We can observe that the solution on the two grids is very similar, showing some slight differences near the boundary of the domain.

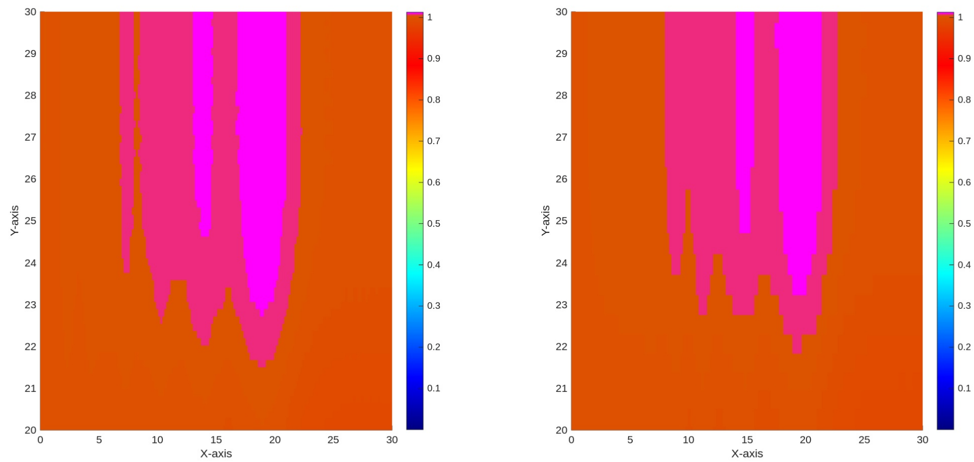

Figure 3.5.: Zoom near the boundary of the domain.

In this zoomed view near the boudary of the domain, we can see that the solution on the non-uniform grid appears to be slightly less accurate, due to the coarser discretization in that region. However, the differences are minimal, and do not significantly affect the overall quality of the solution.

Zooming into the center of the domain, again, no significant differences are observed. However, the refinement of the grid towards the origin in the non-uniform construction is clearly visible.

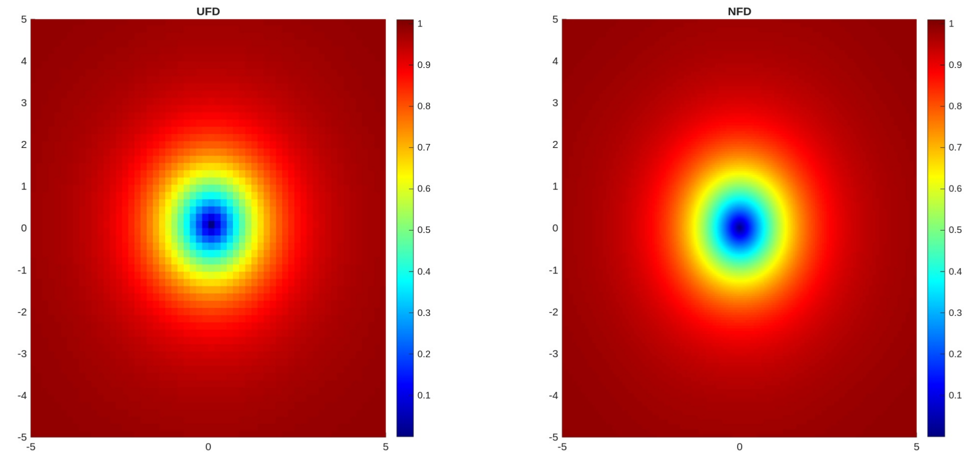

Figure 3.6.: Zoom near the center of the domain.

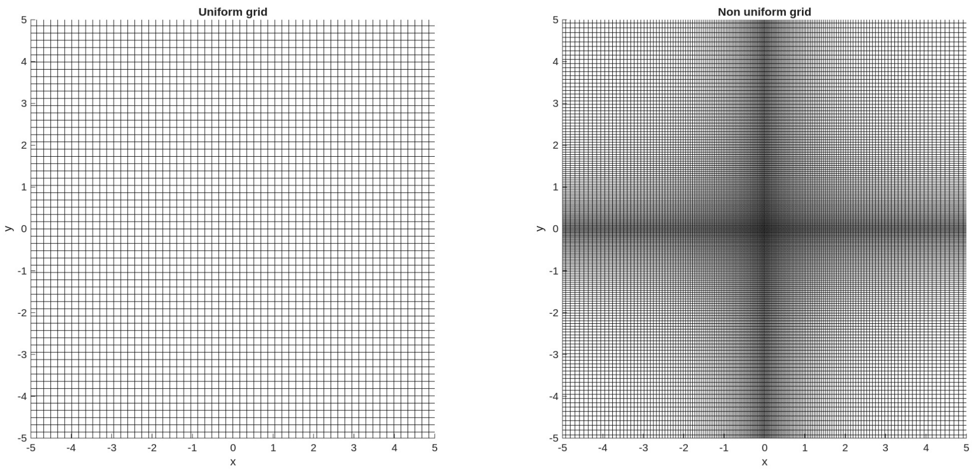

Figure 3.7.: Grid comparison

3.2. Simulations of two straight vortices

In this section, we simulate the dynamics of two interacting straight vortices. We perform experiments by imposing both concordant and opposite initial phases, and compare the different spatial discretization methods introduced so far.

3.2.1. Simulation of marching vortices

We now proceed to perform a set of simulations involving two straight vortices, centered respectively at (x_{c_1}, y_{c_1}) and (x_{c_2}, y_{c_2}). In this case we impose counter-rotating phases, meaning that, given the initial condition

$$\psi_0 = \sqrt{\rho_1\rho_2}\,\exp\big(\mathrm{i}(S_1 + S_2)\big),$$

we have $\rho_1 = \rho(x - x_{c_1}, y - y_{c_1})$ and $S_1 = \operatorname{atan2}(y - y_{c_1}, x - x_{c_1})$, while $\rho_2 = \rho(x - x_{c_2}, y - y_{c_2})$ and $S_2 = -\operatorname{atan2}(y - y_{c_2}, x - x_{c_2})$.

All the following experiments have been carried out under the assumption of symmetry with respect to the y-axis, so that $x_c = x_{c_1} = -x_{c_2}$.

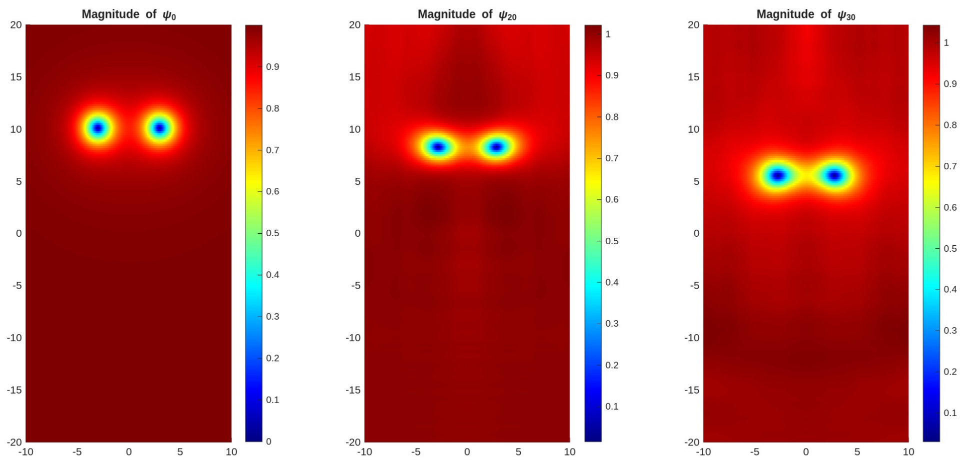

Figure 3.8.: Comparison between $|\psi_0|$ (left), $|\psi_{10}|$ (center), and $|\psi_{30}|$ (right), for $x_c = 3$, $m_x = 259$, $m_y = 201$, and number of time steps $n_t = 1500$.

With this configuration, the vortices appear to march steadily towards the lateral boundaries of the computational domain. This drift is a direct consequence of the imposed counter-rotation, which breaks the otherwise stationary balance. In Figure 3.9 we can appreciate how the translation speed depends strongly on the initial separation between the vortex centers.

All simulations reported here were performed on a nonuniform grid along the x-direction, reported in Figure 3.13 with the spacing defined in Section 2.1.2, while in the y-direction equispaced nodes were employed.

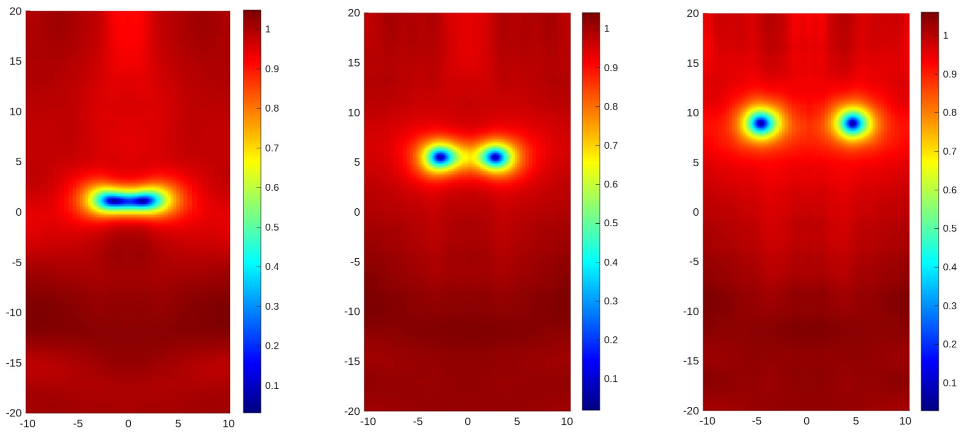

Figure 3.9.: Solutions at time $t = 30$ for $x_c = 2, 3$, and 5.

3.2.2. Simulation of two rotating vortex

We now consider the case of two vortices with the same rotation direction, which results in a rotation of the vortex pair around their center of mass.

The initial condition is given by

$$\psi_0 = \sqrt{\rho_1 \rho_2} \exp\left(\mathrm{i}(S_1 - S_2)\right),$$

with $\rho_1 = \rho(x - x_{c_1}, y - y_{c_1})$ and $S_1 = \operatorname{atan2}(y - y_{c_1}, x - x_{c_1})$, while $\rho_2 = \rho(x - x_{c_2}, y - y_{c_2})$ and $S_2 = \operatorname{atan2}(y - y_{c_2}, x - x_{c_2})$.

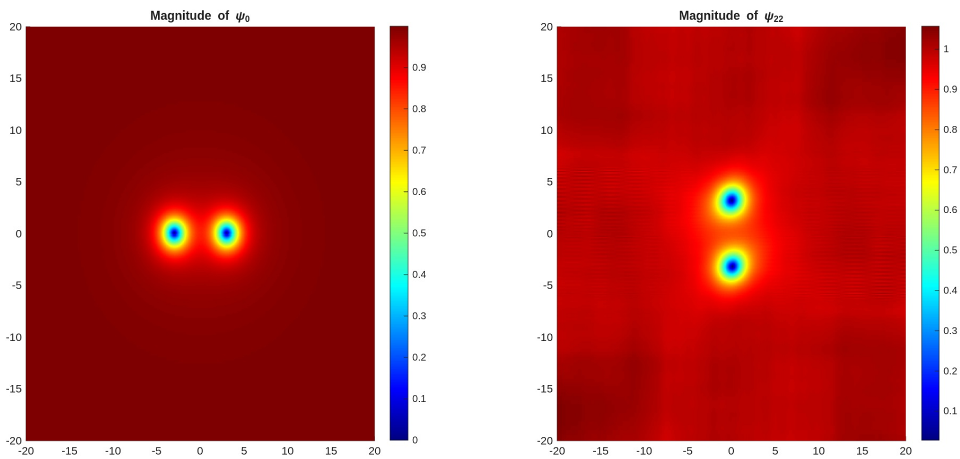

Figure 3.10.: Initial solution (left) and solution at time $t = 22$ (right).

In this simulation, the grid was constructed according to 2.1.2 in both directions, with parameters $x_c = 3$, $h_1 = 0.05$, and $\delta = 0.025$, resulting in $m_x = m_y = 255$ grid points. The time step was set to $\tau = 0.05$.

3.2.3. Comparison between uniform and non-uniform grids

It is particularly interesting to examine how the solution changes depending on the chosen grid, while keeping the parameters fixed. We shall carry out one final experiment to compare the two grids, since in the non-stationary case we observed that the differences are much more pronounced than in the stationary one.

The following experiment was carried out using a nonuniform grid, advancing the simulation up to the time when the two vortices complete approximately a $\pi/2$ rotation. We then compare the vortex positions obtained on the uniform and nonuniform discretizations.

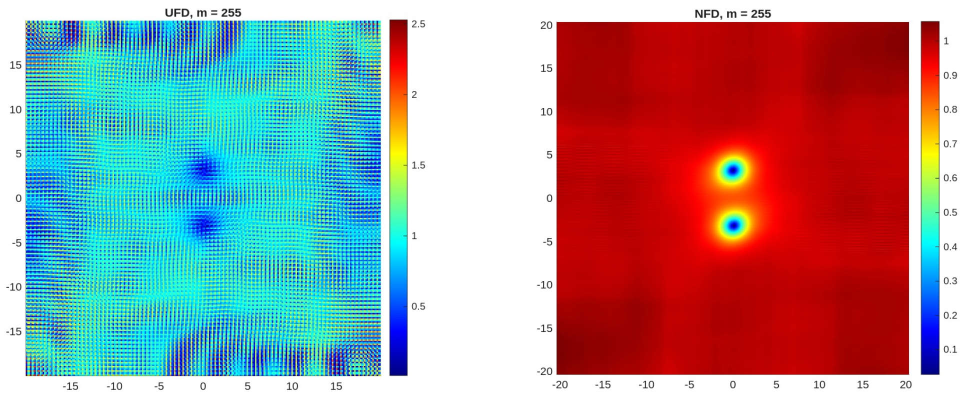

Figure 3.11.: Comparison between equispaced grid and grid as in Figure 3.14

It is clearly visible that the solution obtained with the nonuniform grid is significantly more accurate than the one computed on a uniform grid of the same size.

In order to achieve a comparable level of accuracy with a uniform grid, it is necessary to substantially increase the number of points, reaching as high as $m_x = m_y = 771$, resulting, as well as a much larger computational time.

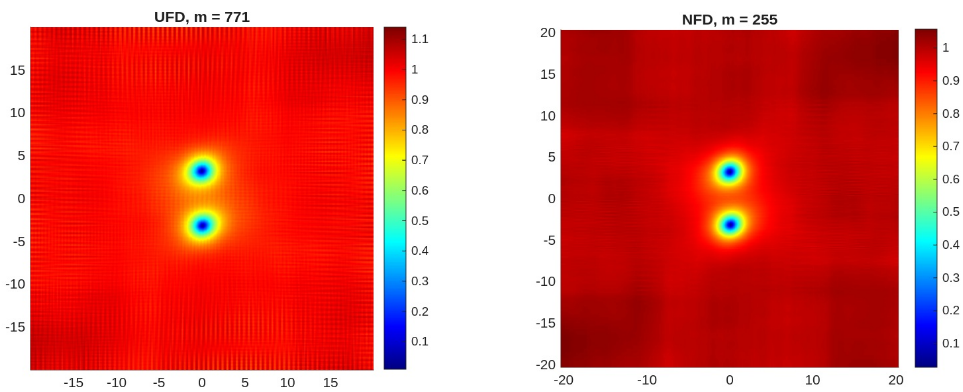

Figure 3.12.: Comparison between uniform grid with $m = 771$ (left) and non-uniform grid with $m = 255$ (right)

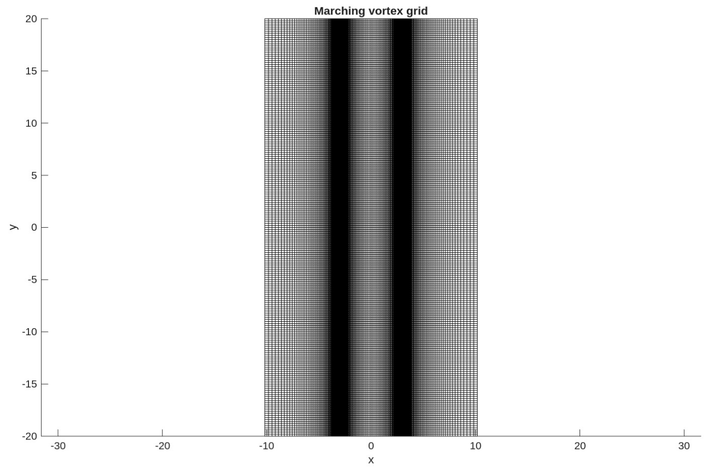

Figure 3.13.: Grid used for the marching vortex simulations

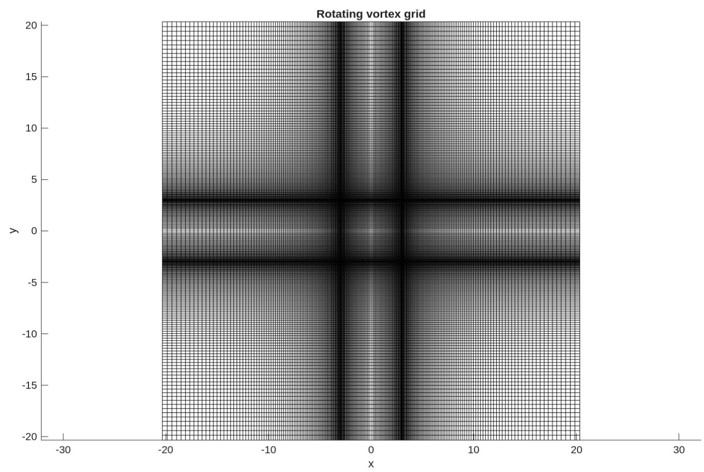

Figure 3.14.: Grid used for rotating vortex simulation

Chapter 4

Conclusions

In this thesis, we investigated the numerical simulation of vortex dynamics in a superfluid system, focusing on the implementation and comparison of central finite-difference schemes on both uniform and non-uniform grids. The time evolution was performed through the Strang splitting method, which proved to be an efficient and reliable approach for the integration of nonlinear partial differential equations of this type.

The proposed construction of the uniform grid turned out to be particularly effective, as it allows the use of large computational domains while maintaining high resolution in the regions of interest, without an unreasonable increase in the total number of grid points. The method has shown good stability and accuracy properties, while keeping computational costs at a reasonable level.

Nevertheless, the approach still suffers from the truncation of the physical domain and the imposition of artificial boundary conditions, which can introduce undesired reflections or distortions near the boundaries. A possible future development of this work would be the implementation of a *free-boundary* method, for instance by rescaling the unbounded domain $\mathbb{R}^2$ into the finite interval $(-1, 1)^2$, thereby eliminating the need for explicit boundary conditions.

Finally, the dynamics of rectilinear vortices were analyzed, showing how, depending on their initial phases, the vortices may either drift towards the domain boundaries or rotate around their common center of mass. The simulations highlighted how both the translational and rotational velocities strongly depend on the initial separation between the vortex cores. This confirms that the implemented numerical framework is capable of capturing the main qualitative and quantitative features of vortex interactions in superfluid systems.

Overall, the results obtained provide a solid foundation for further developments of the numerical model, and represent a step towards a more flexible and accurate computational framework for the study of quantum vortex dynamics.

Appendix A

Kronecker product and some properties

From now on, in this section I denotes the identity matrix.

Definition A.1. Let $A \in \mathbb{C}^{m\times n}$ and $B \in \mathbb{C}^{p\times q}$. The *Kronecker product* is defined as

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix} \in \mathbb{C}^{mp\times nq}.$$

Example A.2.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \otimes \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} = \begin{bmatrix} 0 & 5 & 0 & 10 \\ 6 & 7 & 12 & 14 \\ 0 & 15 & 0 & 20 \\ 18 & 21 & 24 & 28 \end{bmatrix}.$$

Proposition A.3. *If A, B, C, D are matrices of compatible dimensions, then*

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD).$$

Proof. The (i, j) block of $(A \otimes B)(C \otimes D)$ is

$$\sum_k (a_{ik}B)(c_{kj}D) = \Big(\sum_k a_{ik}c_{kj}\Big)BD = (AC)_{ij}BD,$$

which coincides with the (i, j) block of $(AC) \otimes (BD)$. ■

Proposition A.4. *Let $A \in \mathbb{C}^{m\times m}$, $B \in \mathbb{C}^{n\times n}$ and $\mathbf{u} \in \mathbb{C}^{mn}$. Then*

$$(A \otimes B)\mathbf{u} = \operatorname{vec}\left(BUA^T\right)$$

where $\mathbf{u} = \operatorname{vec}(U)$, $U_{i,j} = u_{i+m(j-1)}$

Proof. Consider

$$(A \otimes B)_{(i-1)m+k,\ (j-1)m+\ell} = a_{i,j}\, b_{k,\ell},$$

where $1 \le i, j \le n$ and $1 \le k, \ell \le m$.

The component of the product $(A \otimes B)\,\mathbf{u}$ corresponding to row index $(i-1)\,m + k$ is, by definition,

$$
\begin{aligned}
((A \otimes B)\,\mathbf{u})_{(i-1)m+k} &= \sum_{j=1}^{n} \sum_{\ell=1}^{m} (A \otimes B)_{(i-1)m+k,\,(j-1)m+\ell}\;\; u_{(j-1)m+\ell} \\
&= \sum_{j=1}^{n} \sum_{\ell=1}^{m} a_{i,j}\, b_{k,\ell}\, U_{\ell,j},
\end{aligned}
$$

Reordering the sums and factoring out $a_{i,j}$ gives

$$
((A \otimes B)\,\mathbf{u})_{(i-1)m+k} = \sum_{\ell=1}^{m} b_{k,\ell} \left(\sum_{j=1}^{n} U_{\ell,j}\, a_{i,j} \right).
$$

The inner sum is exactly the (ℓ, i)-entry of the matrix UA^T, since

$$
\left(UA^T\right)_{\ell,i} = \sum_{j=1}^{n} U_{\ell,j} \, \left(A^T\right)_{j,i} = \sum_{j=1}^{n} U_{\ell,j}\, a_{i,j}.
$$

Therefore,

$$
((A \otimes B)\,\mathbf{u})_{(i-1)m+k} = \sum_{\ell=1}^{m} b_{k,\ell} \, \left(UA^T\right)_{\ell,i} = \left(B\,UA^T\right)_{k,i}.
$$

Since we haven't imposed nothing about i and k, this holds for all $i = 1, \ldots, n$ and $k = 1, \ldots, m$, so the claim follows.

■

Lemma A.5. *For every $n \in \mathbb{N}$, one has*

$$
(I \otimes A)^n = I \otimes A^n.
$$

Proof. By induction on n. For $n = 1$ the claim is trivial. Suppose it holds for $n - 1$; then

$$
(I \otimes A)^n = (I \otimes A)(I \otimes A)^{n-1} = (I \otimes A)(I \otimes A^{n-1}) = I \cdot I \otimes A \cdot A^{n-1} = I \otimes A^n
$$

where we used A.3. ■

Lemma A.6. *Let $A, B \in \mathbb{C}^{m \times n}$ and X a matrix. Then $X \otimes A + X \otimes B = X \otimes (A + B)$.*

Proposition A.7. *Let $\{A_k\}_{k=1}^{\infty}$ be a sequence of matrices, where $A_j \in \mathbb{C}^{m \times n}$ for all j. Let I be the identity matrix of arbitrary dimensions. Then*

$$
\sum_{k=1}^{\infty} (I_k \otimes A_k) = I_k \otimes \sum_{k=1}^{\infty} A_k
$$

Proof. We consider the infinite sum as the limit of a partial sum as $\sum_{k=1}^{\infty} (I \otimes A_k) = \lim_{n \to \infty} \sum_{k=1}^{n} (I \otimes A_k)$.

We now proceed for induction on n. For $n = 1$ the claim is trivial. Suppose it holds for $n - 1$, then

$$
\sum_{k=1}^{n} (I \otimes A_k) = I \otimes A_n + \sum_{k=1}^{n-1} (I \otimes A_k) = I \otimes A_n + I \otimes \sum_{k=1}^{n-1} A_k = I \otimes \sum_{k=1}^{n} A_k
$$

■

Theorem A.8. *Let f be an analytic function. Then*

$$f(I \otimes A) = I \otimes f(A) \quad \textit{and} \quad f(A \otimes I) = f(A) \otimes I$$

Proof. Write $f(X) = \sum_{k=0}^{\infty} \alpha_k X^k$. Then

$$f(I \otimes A) = \sum_{k=0}^{\infty} \alpha_k (I \otimes A)^k = \sum_{k=0}^{\infty} \alpha_k (I \otimes A^k) = I \otimes \left(\sum_{k=0}^{\infty} \alpha_k A^k \right) = I \otimes f(A).$$

We proceed in the same way for the other. ■

Appendix B

MATLAB® implementation

In this appendix, we briefly report the implementation of the main function used during the simulations in Chapter 3 in MATLAB®.

The full code is available at the following link[1]. Here we report the main functions and implementation details.

Non uniform grid generation

The function `neqdeltagrid0.m` generates a non-uniform grid as described in Section 2.1.2. The function takes as input the parameters δ, h_1, L and m, and returns the grid points in the interval $[0, L]$.

```
% from xc to 0
for i = m-1:-1:2
    h(i) = (1 + delta)*h(i+1);
    x(i) = x(i+1) - h(i);
end

% from xc to xL
i = m + 1;
while x(i - 1) < xL
    h(i) = (1 + delta)*h(i-1);
    x(i) = x(i-1) + h(i);
    i = i + 1;
end
```

```
% always add 0
x = [0, x];
h = diff(x);

% check for the nearest point to
  0
if numel(x) > 2
    if h(1) < 0.7*h(2)
        x(2) = [];
        h = diff(x);
    end
end
```

We then construct the full grid in $[-L, L]$ by mirroring the points around zero as

```
x = neqdeltagrid0(xc,Lx, delta, h1);
x = unique([-flip(x), x]); x = x(:);
hx = diff(x);
mx = length(x);
```

Once we have the grid points, we can construct the finite difference matrix as

```
%% matrix
dux = 2./(hx(1:mx-2) .* (hx(1:mx-2) + hx(2:mx-1))); % upper diag
dmx = -2./(hx(1:mx-2).*hx(2:mx-1));                  % main diag
dlx = 2./(hx(2:mx-1).*(hx(1:mx-2) + hx(2:mx-1)));   % lower diag

Dxx = spdiags([[dux;0;0], [0; dmx; 0], [0;0;dlx]], -1:1, mx,mx);
```

[1]https://github.com/fofe1125/tesi

```
% Homogeneous Neumann conditions
Dxx(1,1:2) = [-2,2]/(hx(1)^2);
Dxx(mx,mx-1:mx) = [2,-2]/(hx(mx-1)^2);
```

Time splitting

We report a brief section of the code used to implement the time integration

```
for i = 1:nt - 1

    U = Ex*U*Ey.';
    U = exp(1i*tau/2*(1 - abs(U).^2)).*U;
    U = Ex*U*Ey.';

end
```

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